

Scalp

5 Layers (Mnemonic: SCALP)

- **Skin:** The outermost layer of the scalp, which is thick and hairy, containing sebaceous glands and a rich blood supply. It is connected to the epicranial aponeurosis through the superficial fascia.
- **Superficial fascia:** A fibrous and dense layer that binds the skin to the aponeurosis and allows the passage of blood vessels and nerves.
- **Deep fascia (Epicranial aponeurosis / Galea aponeurotica):** A movable layer over the pericranium that receives the occipitofrontalis muscle.

Attachments of Epicranial Aponeurosis

- **Frontalis:** The muscle that is inserted anteriorly into the epicranial aponeurosis, responsible for raising the eyebrows.
- **Occipitalis:** The muscle attached posteriorly to the external occipital protuberance and highest nuchal lines, aiding in pulling the scalp back.
- **Superior temporal line:** A bony ridge where the epicranial aponeurosis is attached laterally, extending to the zygomatic arch.
- **Loose areolar tissue:** This layer extends from the eyelids to the nuchal lines and allows free movement of the upper three layers of the scalp.
- **Pericranium:** A layer loosely attached to the skull bones but firmly adherent at the sutures, forming the outer layer of the skull bones.

Arterial Supply of Scalp

- **Supratrochlear artery:** An artery from the ophthalmic artery that branches off the internal carotid artery, supplying the front of the scalp.
- **Supraorbital artery:** Another artery from the ophthalmic artery supplying the forehead and scalp.
- **Superficial temporal artery:** An artery from the external carotid artery that supplies the side of the head.
- **Posterior auricular artery:** An artery that branches off the external carotid artery, supplying behind the auricle.
- **Occipital artery:** An artery that supplies the posterior aspect of the scalp, originating from the external carotid artery.

Lymphatic Drainage

- **Preauricular / Parotid lymph nodes:** These nodes drain lymph from the anterior part of the scalp.
- **Posterior auricular / Mastoid / Occipital lymph nodes:** These nodes drain lymph from the posterior part of the scalp.

Nerve Supply of Scalp

- **Supratrochlear nerve:** A sensory nerve supplying the forehead region.
- **Supraorbital nerve:** A sensory nerve supplying the forehead and scalp area.
- **Auriculotemporal nerve:** A sensory nerve that supplies the temporal region and part of the auricle.
- **Lesser occipital nerve:** A sensory nerve supplying the skin behind the ear.
- **Greater occipital nerve:** A sensory nerve that supplies the posterior scalp.
- **Third occipital nerve:** A sensory nerve providing sensation to the lower scalp area.
- **Temporal branch of facial nerve:** A motor nerve that innervates the muscles of facial expression in the temple area.
- **Posterior auricular branch of facial nerve:** A motor nerve innervating the muscles around the ear.

- **Sensory Nerves:** Nerves that provide sensation, including the supratrochlear, supraorbital, zygomaticotemporal, auriculotemporal, great auricular, lesser occipital, greater occipital, and third occipital nerves.
- **Motor Nerve:** Nerves that provide movement, including the temporal branch of the facial nerve (in front of the auricle) and the posterior auricular branch of the facial nerve (behind the auricle).

- **Cervical plexus:** A network of nerves in the neck.
- **Occipital:** Related to the back of the head.
- **Dorsal ramus:** A branch of a spinal nerve supplying the back of the body.
- **C3 (Cervical vertebra 3):** The third vertebra in the neck.
- **Facial nerve:** A nerve that controls facial expressions.

Trigeminal Nerve (CN V)

Origin: The point where something begins or arises.

Mixed Nerve: A nerve that contains both sensory (afferent) and motor (efferent) fibers.

Branches: The divisions or offshoots of a nerve.

Ophthalmic: Related to the eye.

Maxilla: The upper jawbone.

Mandibular: Related to the lower jaw.

Sensory: Pertaining to the perception of sensations (like touch, pain).

Motor: Related to movement or muscular activity.

Mastication: The process of chewing.

Pathway: The route taken by nerve fibers.

Trigeminal Ganglion: A cluster of nerve cell bodies associated with the trigeminal nerve.

Middle Cranial Fossa: A depression in the base of the skull that houses parts of the brain.

Reflexes: Automatic responses to stimuli.

Corneal Reflex: An involuntary blink response when the cornea is touched.

Jaw Jerk Reflex: A reflex that involves the rapid closure of the jaw when the chin is tapped.

Maxillary Nerve (V2) - Sensory Nerve

Origin: The point where something begins or arises.

Foramen Rotundum: A round opening in the skull through which the maxillary nerve exits the cranial cavity.

Pterygopalatine Fossa: A space in the skull behind the maxilla that contains nerves and blood vessels.

Pathway: The route taken by the nerve fibers.

Infraorbital Nerve: A branch of the maxillary nerve located in the floor of the orbit.

Infraorbital Canal: A tunnel within the maxilla through which the infraorbital nerve and artery travel.

Infraorbital Foramen: An opening in the maxilla through which the infraorbital nerve exits to the face.

Peripheral Branches: The smaller branches of the main nerve that supply specific areas.

Inferior Palpebral Branch: A branch supplying the lower eyelid and conjunctiva.

Lateral Nasal Branch: A branch supplying the lateral skin of the external nose.

Superior Labial Branch: A branch supplying the skin and mucosa of the upper lip and cheek.

Branches of Maxillary Nerve

Meningeal Branches: Nerve branches that supply the dura mater of the middle cranial fossa.

Ganglionic Branches: Branches that join the pterygopalatine ganglion, carrying sensory and secretomotor fibers.

Zygomatic Nerve: A branch of the maxillary nerve that supplies the zygomatic region.

Zygomaticotemporal: A branch that supplies the temporal region and scalp and carries fibers to the lacrimal gland.

Zygomaticofacial: A branch that supplies the skin of the malar region (cheek area).

Posterior Superior Alveolar Nerve: A nerve supplying the maxillary air sinus, molar teeth, and maxillary gum.

Middle Superior Alveolar Nerve: A nerve supplying the premolar teeth that may join the superior dental plexus.

Anterior Superior Alveolar Nerve: A nerve supplying the incisor and canine teeth and forming the superior dental plexus with other alveolar nerves.

Ganglionic Branches

Sensory Fibers: Nerve fibers that carry sensory information from various regions.

Secretomotor Fibers: Nerve fibers that stimulate secretion from glands.

Nerve of Pterygoid Canal: A nerve that carries secretomotor fibers to the lacrimal gland.

Zygomatic Nerve

Branch of Maxillary Nerve: A division of the maxillary nerve that supplies the zygomatic area.

Malar Region: The area of the cheek.

Superior Alveolar Nerves

Posterior Superior Alveolar Nerve: A nerve that supplies the maxillary air sinus, molar teeth, and gum.

Middle Superior Alveolar Nerve: A nerve supplying the premolars and part of the superior dental plexus.

Anterior Superior Alveolar Nerve: A nerve that supplies the incisors, canines, and nasal wall.

Applied Anatomy

Trigeminal Neuralgia: A condition characterized by severe pain along the trigeminal nerve, which includes the maxillary nerve.

Cavernous Sinus Thrombosis: A condition where a blood clot in the cavernous sinus may compress the maxillary nerve, causing symptoms.

Mandibular Nerve

Largest Division: The biggest branch of the trigeminal nerve.

Mixed Nerve: A nerve that contains both sensory (afferent) and motor (efferent) roots.

Sensory Root: The part of the nerve that carries sensory information, originating from the trigeminal ganglion.

Motor Root: The part of the nerve responsible for motor functions, located deep to the trigeminal ganglion.

Course

Foramen Ovale: An oval-shaped opening in the skull through which the mandibular nerve passes.

Infratemporal Fossa: A space located below the base of the skull, behind the maxilla.

Mandibular Nerve Trunk: The main trunk formed by the unification of the sensory and motor roots.

Tensor Veli Palatini: A muscle of the soft palate.

Lateral Pterygoid: A muscle involved in the movement of the jaw.

Otic Ganglion: A cluster of nerve cell bodies located on the medial side of the mandibular nerve trunk.

Anterior Division: The branch of the mandibular nerve that is mainly motor.

Posterior Division: The branch of the mandibular nerve that is mainly sensory.

Branches

Main Trunk: The primary pathway from which branches of the mandibular nerve arise.

Meningeal Branch (Nervous Spinosus): A branch that enters the foramen spinosum and supplies the dura mater.

Medial Pterygoid: A muscle involved in jaw movement.

Tensor Tympani: A muscle in the middle ear that helps to dampen sound.

From Anterior Division (Mainly Motor)

Nerve to Masseter: A branch supplying the masseter muscle and the temporomandibular joint (TMJ).

Nerve to Lateral Pterygoid: A branch supplying the lateral pterygoid muscle.

Deep Temporal Nerves: Nerves that supply the temporalis muscle.

Buccal Nerve: A sensory nerve supplying the cheek mucosa and skin.

From Posterior Division (Mainly Sensory)

Auriculotemporal Nerve: A nerve with branches supplying various areas, including:

- **Auricular Branches:** Supply the external ear.
- **Temporal Branches:** Supply the skin of the temporal region.
- **Glandular Branches:** Supply the parotid gland.
- **TMJ Branch:** Supplies the temporomandibular joint.

Inferior Alveolar Nerve: A mixed nerve that enters the mandibular foramen and supplies the pulp and periodontium of the mandibular teeth.

Nerve to Mylohyoid: A motor nerve supplying the mylohyoid muscle and the anterior belly of the digastric muscle.

Inferior Dental Plexus: A network of nerves formed by branches supplying the molars and premolars.

Incisive Branch: A branch that supplies the incisors and canine teeth.

Mental Nerve: A branch that emerges from the mental foramen and supplies the lower lip and chin.

Lingual Nerve (Sensory)

Joined by Chorda Tympani: The lingual nerve is connected to a branch of the facial nerve.

Mandibular Third Molar: The lower wisdom tooth.

Hyoglossus Muscle: A muscle of the tongue.

Submandibular Duct: A duct that carries saliva from the submandibular gland.

Supplies Anterior 2/3rd of Tongue: Provides sensation to the front two-thirds of the tongue, the floor of the mouth, and the mandibular gums.

Facial Artery

Main Artery: The principal blood vessel supplying the face, located within the muscles responsible for facial expressions.

Tortuous Course: A winding or curvy path taken by the artery as it traverses the face.

Origin

Arises: The point at which the artery begins or comes from.

External Carotid Artery: A major artery in the neck that supplies blood to the face and neck.

Greater Horn of Hyoid Bone: The larger part of the hyoid bone located in the neck, serving as a point of reference for the artery's origin.

Carotid Triangle: An anatomical region in the neck where the common carotid artery bifurcates into the internal and external carotid arteries.

Course

Submandibular Gland: A salivary gland located beneath the mandible (lower jaw).

Stylomandibular Ligament: A ligament that connects the styloid process of the temporal bone to the mandible.

Deep Cervical Fascia: A connective tissue layer in the neck that surrounds and supports structures.

Anteroinferior Angle of Masseter: The front lower corner of the masseter muscle, a major muscle used in chewing.

Anaesthetist's Artery: A colloquial term for the facial artery due to its location and significance in dental anesthesia.

Ascends: To move or go upward.

Medial Angle of Eye: The inner corner of the eye where the eyelids meet.

Lacrimal Sac: A structure that collects tears from the eye before they drain into the nasal cavity.

Anastomosing: The process by which blood vessels connect with each other.

Branches in Neck

Ascending Palatine Artery: A branch of the facial artery that supplies blood to the palate.

Tonsillar Artery: A branch that supplies blood to the palatine tonsil.

Glandular Branches: Branches of the artery that supply blood to glands, specifically the submandibular gland and nearby lymph nodes.

Submental Artery: A branch that supplies the area beneath the chin.

Branches on Face

Inferior Labial Artery: A branch that supplies blood to the lower lip.

Superior Labial Artery: A branch that supplies blood to the upper lip and nose.

Lateral Nasal Artery: A branch that supplies the lateral aspect of the nose.

Muscular Branches: Branches that supply blood to the muscles responsible for facial expressions.

Applied Anatomy

Facial Artery Compression: Applying pressure to the artery against the lower border of the mandible to control or stop bleeding.

Labial Branches: Arterial branches supplying the lips, positioned closer to the mucous membrane than the skin.

Tooth Injuries: Damage to teeth that can lead to internal bleeding (hematomas) in the lips.

Labial Arteries Anastomose: The connection between labial arteries on both sides of the mouth, allowing for blood flow between the external carotid arteries of both sides.

Course of Facial Nerve

Origin

Pons: A part of the brainstem located above the medulla and below the midbrain, playing a crucial role in relaying signals.

Pontomedullary Junction: The point where the pons meets the medulla oblongata.

Motor Root: The branch of the facial nerve responsible for controlling the muscles involved in facial expressions.

Nervus Intermedius: A branch of the facial nerve that carries taste and secretomotor fibers.

Internal Acoustic Meatus: A narrow canal in the temporal bone through which the facial and vestibulocochlear nerves pass.

Petrous Part: The dense, rock-like portion of the temporal bone housing the inner ear.

Intracranial Course

Facial Canal: A bony canal within the temporal bone that houses the facial nerve.

Geniculate Ganglion: A collection of nerve cell bodies located at the first bend of the facial canal, involved in sensory and autonomic functions.

Greater Petrosal Nerve: A branch of the facial nerve supplying the lacrimal gland and nasal mucosa.

Nerve to Stapedius: A small branch that innervates the stapedius muscle, which helps control sound vibrations.

Chorda Tympani: A branch that carries taste sensations from the anterior two-thirds of the tongue and secretomotor fibers to certain salivary glands.

Stylomastoid Foramen: An opening in the temporal bone where the facial nerve exits the skull.

Extracranial Course

Emerges: To come out or become visible.

Stylomastoid Foramen: The opening where the facial nerve exits the skull, located between the styloid and mastoid processes of the temporal bone.

Posterior Auricular Nerve: A branch of the facial nerve that supplies the muscles of the external ear and the occipitalis muscle.

Digastric Muscle: A muscle with two muscle bellies that helps in depressing the mandible.

Stylohyoid Muscle: A muscle that elevates the hyoid bone during swallowing.

Parotid Gland: The largest salivary gland located near the ear.

Terminal Branches: The final divisions of a nerve that provide innervation to specific areas or muscles.

Functions of the Facial Nerve

Motor: Related to the control of muscles.

Secretomotor: Referring to the nerve's role in stimulating glandular secretions.

Taste: The sensation associated with the perception of flavors.

Sensory: Pertaining to the nerve's role in transmitting sensory information.

Applied Anatomy

Bell's Palsy: A condition causing sudden, temporary paralysis of the facial muscles, typically due to inflammation of the facial nerve.

Crocodile Tears Syndrome: A phenomenon where the miswiring of nerve fibers causes tearing during eating, resembling a crocodile's tears.

Ramsay Hunt Syndrome: A condition caused by a viral infection affecting the geniculate ganglion, leading to facial paralysis and ear pain.

Lacrimal Gland

Serous Gland: A type of gland that produces a watery secretion, such as tears.

'J' Shaped: Describes the anatomical shape of the lacrimal gland, resembling the letter "J."

Indented: Referring to being marked or shaped by an external structure, such as the tendon of a muscle.

Levator Palpebrae Superioris Muscle: A muscle responsible for elevating the upper eyelid.

Lacrimal Fossa: A small depression in the bony orbit that houses the lacrimal gland.

Bony Orbit: The bony cavity that contains the eyeball and its associated structures.

Conjunctival Fornices: The spaces between the eyelids and the eyeball where the conjunctiva forms a pocket.

Parts of the Lacrimal Gland

Palpebral Part: The smaller, more superficial section of the lacrimal gland located within the eyelid.

Orbital Part: The larger, deeper section of the lacrimal gland situated in the orbit.

Ducts

Ducts: Tubes that carry secretions from the gland to the conjunctival sac.

Conjunctival Sac: The space formed by the conjunctiva, which covers the front of the eye and the inner surfaces of the eyelids.

Superior Fornix: The upper recess of the conjunctival sac where the upper eyelid meets the eyeball.

Blood Supply and Innervation

Lacrimal Branch: A branch of the ophthalmic artery that supplies blood to the lacrimal gland.

Ophthalmic Artery: An artery that supplies blood to the eye and its associated structures.

Innervated: Supplied with nerves that provide sensation and control functions.

Lacrimal Nerve: A branch of the ophthalmic nerve that carries sensory and secretomotor fibers to the lacrimal gland.

Course of Secretomotor Fibers

Lacrimal Nucleus: A collection of nerve cells in the brain that are involved in tear production.

Nervus Intermedius: A branch of the facial nerve that carries sensory and secretomotor fibers.

Geniculate Ganglion: A collection of nerve cell bodies located along the facial nerve.

Greater Petrosal Nerve: A branch of the facial nerve that carries fibers to the lacrimal gland.

Nerve of Pterygoid Canal: A nerve that carries secretomotor fibers to the pterygopalatine ganglion.

Pterygopalatine Ganglion: A collection of nerve cell bodies that relay secretomotor fibers to the lacrimal gland.

Zygomatic Nerve: A branch that carries fibers to the lacrimal gland.

Zygomaticotemporal Nerve: A branch of the zygomatic nerve that carries fibers to the forehead and temples.

Lacrimal Fluid

Secreted: Released from a gland or organ.

Lubricating: The act of moistening to reduce friction.

Evaporates: Transforms from liquid to vapor, disappearing into the air.

Lacrimal Canaliculi: Small channels that drain excess tears from the conjunctival sac into the lacrimal sac.

Overflow: The process of liquid exceeding its container, leading to spilling, such as tears spilling over when in excess.

Levator Palpebrae Superioris Muscle

Eye Opener Muscle: A term describing the function of the levator palpebrae superioris muscle, which lifts the upper eyelid.

Dilator Muscle: A type of muscle that enlarges or opens a structure, such as the eye opening.

Origin

Inferior Surface: The underside of a structure, in this case, the lesser wing of the sphenoid bone.

Lesser Wing of the Sphenoid Bone: A part of the sphenoid bone located at the base of the skull, contributing to the eye socket.

Orbital Surface: The surface of the bone that forms part of the orbit (eye socket).

Body of the Sphenoid Bone: The central portion of the sphenoid bone, which supports the skull and contains the sphenoidal sinuses.

Anterior to the Optic Foramen: Positioned in front of the opening that allows the optic nerve to pass from the eye to the brain.

Insertion

Expands: The muscle fibers spread out to cover a larger area.

Superficial and Deep Layers: Refers to the two distinct layers into which the muscle divides as it approaches its insertion.

Orbital Septum: A fibrous membrane that separates the orbit from the eyelids.

Palpebral Ligaments: Fibrous structures that connect the eyelids to the surrounding bones of the orbit.

Superior Tarsal Plate: A fibrous structure that provides support to the upper eyelid.

Smooth Muscle Fibers: Muscle cells that are involuntary and not under conscious control, found in the levator palpebrae superioris.

Müller's Muscle: Another name for the smooth muscle fibers within the levator palpebrae superioris that assist in elevating the eyelid.

Aponeurosis: A flat, sheet-like tendon that connects muscles to the parts they move; in this case, it separates the lacrimal gland's components.

Nerve Supply

Upper Division of the Oculomotor Nerve: The branch of the oculomotor nerve responsible for innervating the levator palpebrae superioris muscle.

Sympathetic Fibers: Nerve fibers that originate from the sympathetic nervous system, which controls involuntary bodily functions.

T1 Segment of the Spinal Cord: The first thoracic nerve root, part of the spinal cord that gives rise to sympathetic fibers supplying the smooth muscle.

Actions

Elevates the Upper Eyelid: The primary function of the levator palpebrae superioris muscle, raising the eyelid.

Applied Anatomy

Paralysis: Loss of function or ability to move a muscle.

Ptosis: The medical term for drooping of the upper eyelid.

Smooth Muscle Fibers: In the context of paralysis, these refer to the involuntary muscle fibers that, when affected, can lead to eyelid drooping.

Pseudoptosis: A condition that mimics ptosis, often caused by issues other than direct muscle paralysis.

Horner's Syndrome: A neurological condition resulting from disruption of sympathetic nerve pathways, leading to symptoms such as ptosis and miosis (constricted pupil).

Platysma

Platysma: A thin, flat muscle located in the neck, involved in facial expression.

Pectoral Fascia: A connective tissue layer covering the muscles in the chest area.

Deltoid Fascia: The connective tissue covering the deltoid muscle, located in the shoulder.

Fibers: The elongated cells that make up the muscle tissue.

Base of the Mandible: The lower part of the jawbone.

Skin of the Lower Face and Lip: Refers to the outer layer of the skin covering the lower portion of the face.

Continuous with the Risorius: Indicates that the platysma muscle fibers may blend with the risorius muscle, which helps in facial expressions.

Actions: The functions performed by the muscle.

Releases Pressure: Refers to the muscle's ability to relieve tension in the skin.

Depresses the Mandible: The action of lowering the lower jaw.

Pulls the Angle of the Mouth Downwards: The ability to draw the corners of the mouth down, often indicating emotions like fear or surprise.

Buccinator

Buccinator: A muscle of the cheek that plays a key role in facial expressions and chewing.

Origin: The point where a muscle begins.

Upper Fibers: The topmost muscle fibers that originate from the maxilla (upper jawbone).

Maxilla: The bone that forms the upper jaw and part of the eye socket.

Lower Fibers: The bottom muscle fibers that originate from the mandible (lower jawbone).

Mandible: The bone that forms the lower jaw.

Middle Fibers: The fibers located between the upper and lower fibers that originate from the pterygomandibular raphe.

Pterygomandibular Raphe: A fibrous seam between the pterygoid and mandible that provides attachment for muscles.

Insertion: The point where a muscle attaches to its target structure.

Decussate: To cross over; in this context, it refers to how the middle fibers cross each other before attaching to the lips.

Actions: The functions performed by the muscle.

Flattens Cheek Against Gums and Teeth: The action of the buccinator muscle to keep food from accumulating in the mouth.

Prevents Accumulation of Food in the Vestibule: The role of the buccinator in avoiding food build-up in the space between the gums and inner cheeks.

Facial Vein

Facial Vein: The largest vein in the face that drains blood, lacking valves to prevent backflow.

Valves: Structures within veins that prevent the backward flow of blood.

Begins as the Angular Vein: The facial vein starts from the angular vein located at the inner corner of the eye.

Medial Angle of the Eye: The inner corner where the upper and lower eyelids meet.

Supratrochlear Vein: A vein that drains the forehead area.

Supraorbital Vein: A vein that drains the forehead and upper eyelid region.

Union: The point at which two or more veins come together to form a single vein.

Running Downwards and Backwards: The direction in which the facial vein travels.

Anteroinferior Angle of the Masseter: The front and lower corner of the masseter muscle, which is responsible for chewing.

Pierces the Deep Fascia: The facial vein penetrates the connective tissue surrounding the muscles.

Crosses the Submandibular Gland: The facial vein moves over the gland located beneath the jaw.

Retromandibular Vein: A vein located behind the mandible that collects blood from the face and neck.

Common Facial Vein: The vein formed by the joining of the facial vein and the retromandibular vein.

Internal Jugular Vein: A major vein that drains blood from the head and neck back to the heart.

Emissary Vein

Emissary Vein: Veins that connect intracranial venous sinuses to veins outside the skull.

Intracranial Venous Sinuses: Channels within the cranial cavity that collect blood from the brain.

Extracranial Veins: Veins located outside the skull that drain blood from the face and neck.

Function: The role or purpose of the emissary veins.

Venous Drainage: The process of removing deoxygenated blood from tissues.

Occipital Vein: The emissary vein that connects the occipital sinus (located at the back of the skull) to neck veins.

Parietal Vein: The emissary vein that connects the superior sagittal sinus (a major sinus in the brain) to scalp veins.

Mastoid Vein: The emissary vein that connects the sigmoid sinus (located near the base of the skull) to veins in the ear region.

Clinical Relevance: The importance of emissary veins in medical contexts.

Infections: The invasion of the body by harmful microorganisms.

Cranial Cavity: The space within the skull that houses the brain.

External Environment: The area outside the body that can contain pathogens.

Orbicularis Oculi

Orbicularis Oculi: A muscle that encircles the eye, involved in closing the eyelids.

Composed: Made up of; consisting of.

Parts: The different sections or divisions of a structure.

Orbital Part: The portion of the orbicularis oculi that surrounds the orbit (eye socket).

Origin: The starting point where a muscle attaches to a bone or structure.

Medial Palpebral Ligament: A ligament located at the inner corner of the eyelids.

Adjoining Bone: The bone that is next to or associated with the ligament.

Insertion: The attachment point of the muscle at the opposite end from its origin.

Concentric Rings: Circular layers that surround a common center.

Action: The function or movement produced by the muscle.

Closes Lids Tightly: The ability of the muscle to firmly shut the eyelids.

Protects Eye: The role of the muscle in safeguarding the eye from harmful stimuli.

Palpebral Part: The section of the orbicularis oculi responsible for gentle eyelid closure.

Lateral Part of Medial Palpebral Raphe: The outer edge of the seam between the eyelids.

Blinking: The rapid closing and opening of the eyelids.

Lacrimal Part: The portion of the orbicularis oculi associated with the lacrimal system.

Lacrimal Fascia: A connective tissue layer associated with the lacrimal (tear) gland.

Lacrimal Bone: A small bone located in the eye socket that contributes to tear drainage.

Dilates Lacrimal Sac: The action of expanding the sac that collects tears.

Supports Lower Eyelid: The role of the muscle in maintaining the position of the lower eyelid.

Palpebral Ligament

Palpebral Ligament: A connective tissue structure that helps support the eyelids.

Palpebral Fascia: The connective tissue covering the eyelids.

Orbital Septum: The fibrous tissue that separates the eye from the eyelids.

Thickenings: Areas where tissue becomes denser or more substantial.

Tarsal Plates: The stiff structures that provide support to the eyelids.

Tarsi: Referring to the tarsal plates or the fibrous structures within the eyelids.

Dangerous Area of Scalp

Dangerous Area of Scalp: A specific region of the face where infections can spread easily.

Infections: The presence of harmful microorganisms that can cause disease.

Retrograde Direction: Movement backward or in the opposite direction of normal flow.

Thrombosis: The formation of a blood clot within a blood vessel.

Cavernous Sinus: A large vein located at the base of the skull that drains blood from the brain.

Presence of Infection: The existence of harmful microorganisms in a specific area.

Upper Lip: The top part of the lip, which can be a site of infection.

Lower Part of the Nose: The lower region of the nose, which can also be affected by infections.